

Growth Theory (Part 1)

Intermediate Macroeconomics

Fall 2025, Sciences Po Campus du Havre

Lecture 2

Announcements

- Textbook: Chapter 10.4 and 11.1
- Quiz 1 announced today, due next Tuesday (9/23)

The Facts of Growth

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We start by focusing on the determination of growth in the long run.

What do economists know about growth?

- Empirics: stylized facts about economic growth (today's lecture)
- Theory: growth models (next lectures)

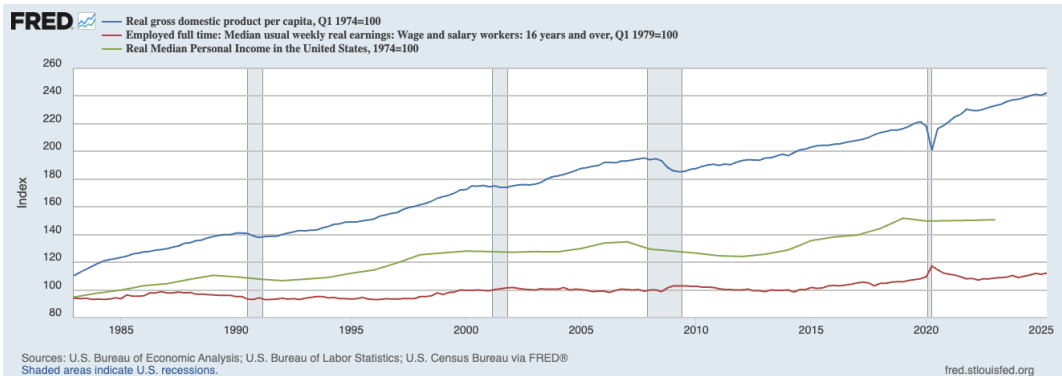
Stylized facts about economic growth

- In the real world, we observe for each country time series of macroeconomic variables such as GDP, consumption, unemployment, etc.
- We also observe certain patterns (correlations, stylized facts) either over time or in the cross-section

Examples:

- Time-series: during booms and recessions, output, consumption, investment and unemployment all move together in the same direction
- Cross-section: richer countries tend to have more capital

US: Output per capita



Measuring the standard of living

We care about growth because we care about the standard of living

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- **Purchasing power parity (PPP)**
- The right measure on the production side is **output per worker** or **output per hour worked**.

Purchasing power parity (PPP)

- PPP is the rate of currency conversion that equalizes the purchasing power of different currencies by eliminating the differences in price levels between countries.
- In the simplest form, PPP shows the ratio of prices in national currencies of the same good or service in different countries.
- PPP is also calculated for a basket of goods and services
- Example of PPP data: [OECD](#)

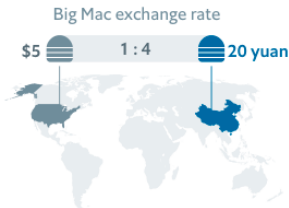
PPP and the Big Mac index

How it works

Purchasing-power parity implies that exchange rates are determined by the value of goods that currencies can buy



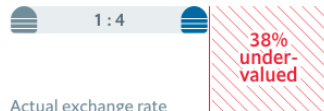
Differences in local prices – in our case, for Big Macs – can suggest what the exchange rate should be



Raw index / GDP-adjusted

Using burgeronomics, we can estimate how much one currency is under- or over-valued relative to another

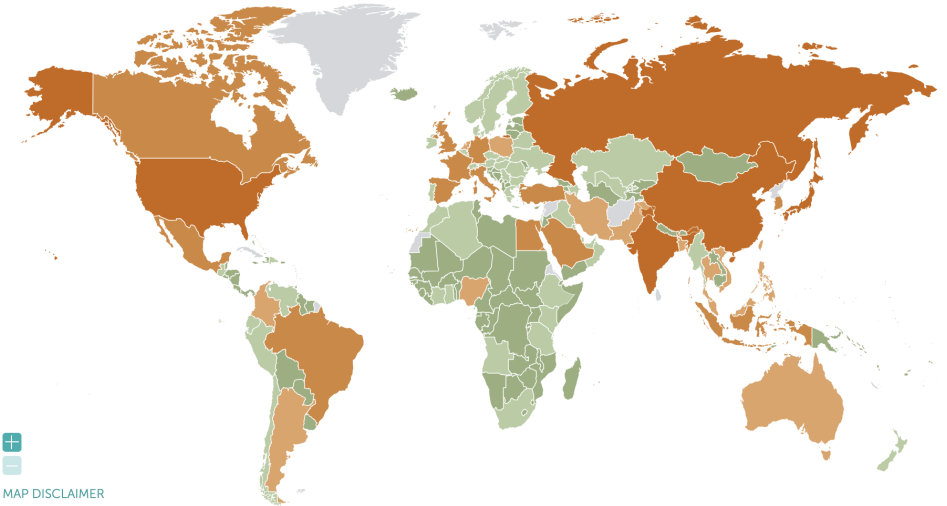
Big Mac exchange rate



Source: the Economist

Distribution of the world GDP, PPP in 2025

Greater than 3% 1% - 3% 0.5% - 1% 0.1% - 0.5% Less than 0.1% no data



MAP DISCLAIMER

Figure: data source as IMF

GDP per capita, PPP

Output per person, rather than output itself, is the variable we compare over time or across countries

● 25,000 or more ● 15,000 - 25,000 ● 5,000 - 15,000 ● 1,000 - 5,000 ● under 1,000 ● no data

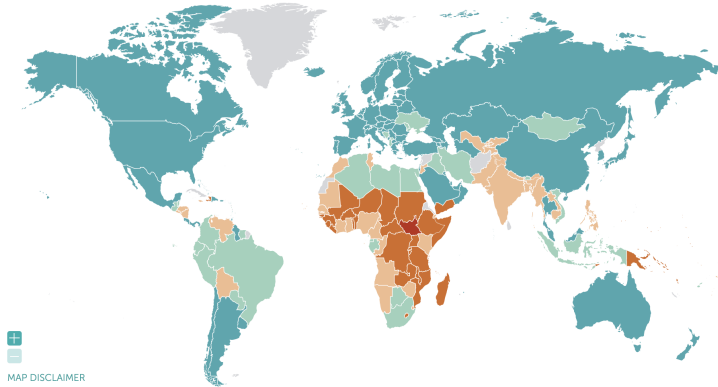


Figure: Data source as IMF

GDP per capita, PPP

TOP 10

- 1 Singapore 156,76K
- 2 Luxembourg 152,92K
- 3 Macao SAR 134,04K
- 4 Ireland 134K
- 5 Qatar, 121,61K
- 6 Norway 107,89K
- 7 Switzerland 97,58K
- 8 Brunei Darussalam 95,76K
- 9 Guyana 94,26K
- 10 USA 89,11K

BOTTOM 10

- Burundi 1.02K
Central African Republic 1.33K
Yemen 1.67K
Mozambique 1.73K
Malawi 1,78K
Congo Dem 1.88K
Somalia 1.92K
Liberia 2.01K
Madagascar 2.04K
Niger 2.1K

Source: IMF, WEO

Growth compounding

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Increase of standard of living: $Y_{10} - Y_0 = 1.02^{10} - 1 = 1.21899 - 1 = 0.21899 = 22\%$

Does Money Lead to Happiness?

Easterlin paradox:

- *Across countries*: higher income correlates with higher GDP, valid only for relatively poor countries. No relationship for rich countries
- *Across time*: average happiness in a given (rich) country does not seem to increase over time, no data on poor countries
- *Across people*: strong correlation between happiness and income

What is the conclusion?

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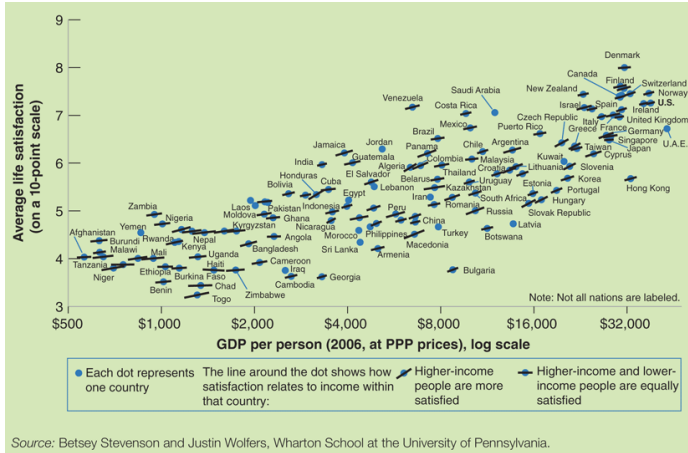
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What is the conclusion?

- Once basic needs are satisfied, higher income per person does not increase happiness
- The level of income *relative to others* (rather than the absolute level of income) matters

Does Money Lead to Happiness?

Stevenson and Wolfers (2008):

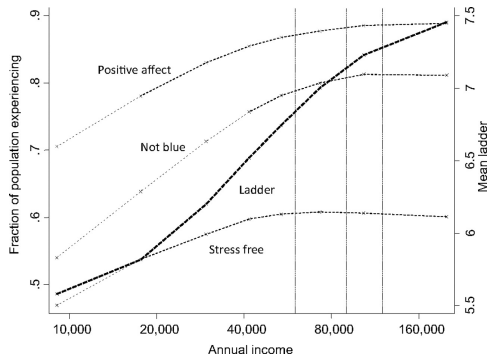


Clear upward relationship between income and happiness

Does Money Lead to Happiness?

Deaton and Kahneman (2010)

- Emotional well-being: frequency and intensity of joy, stress, etc.
- Life satisfaction: personal assessment of your life (ladder)



Income can bring life satisfaction but not necessarily happiness. Money does buy emotional well-being, but only up to a point (est. \$75,000)

Further reading

Barro, R.J, and Sala-i-Martin, X. (2003). Economic growth, 2nd ed. MIT Press, Cambridge.

Kahneman, Daniel and Deaton, Angus (2010): “High income improves evaluation of life but not emotional well-being” , *Proceedings of the National Academy of Sciences*, 107 (38) 16489–16493

IMF (2018), World Economic Outlook, April [link here](#)

Stevenson, Betsay and Wolfers, Justin (2008): Economic Growth and Subjective Well-Being: Reassessing the Easterlin Paradox,

The Economist, The Big Mac Index: [link here](#)

The United Nations (2018), World Happiness Report: [link here](#)

Production Function

Thinking about growth: production function

Aggregate production function F :

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where Y is output, K is capital, and N is labor.

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- The function F depends on the state of technology.
- The most common production function in Cobb-Douglas:

$$Y = K^{\alpha} N^{1-\alpha}$$

where $0 < \alpha < 1$

Properties of the production function

1. Constant returns to scale (CRS)

- If we double **all** inputs $\implies$ the output doubles:

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2. Positive and diminishing marginal products

- **Decreasing returns to capital:** Increases in capital lead to smaller and smaller increases in output.
- **Decreasing returns to labor:** Increases in labor lead to smaller and smaller increases in output.

Output and capital

- Properties of the production function imply a simple relation between: output per worker and capital per worker:

- Textbook notation:

$$\frac{Y}{N} = F\left(\frac{K}{N}, \frac{N}{N}\right) = F\left(\frac{K}{N}, 1\right)$$

- In the lectures we will simplify the notation:
small letter will denote “per worker” variables: $y = Y/N$ and $k = K/N$

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- Increases in capital per worker: Movements along the production function
- Improvements in the state of technology: Shifts (up) of the production function

Growth theory

Output and Capital (advanced version)

Output in the long run depends on two relations:

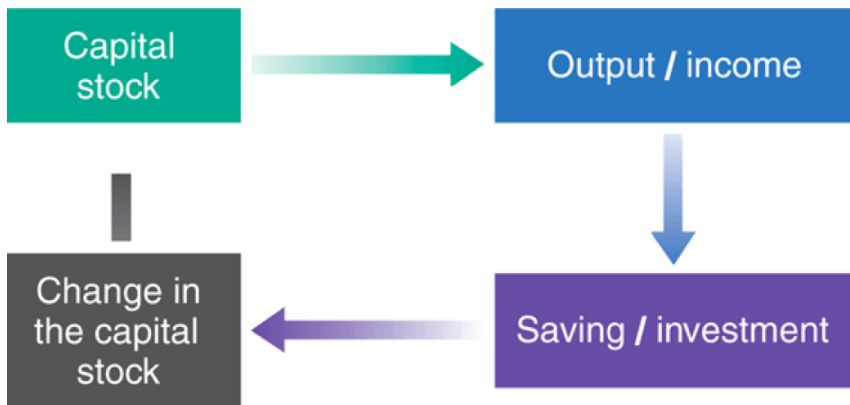
1. **Production function:** the amount of capital determines the amount of output

Output and Capital (advanced version)

Output in the long run depends on two relations:

1. **Production function**: the amount of capital determines the amount of output
2. The amount of output being produced determines the amount of **saving**, which in turn determines the amount of **capital being accumulated over time**

Output, Capital and Saving/Investment



The law of motion of capital

$$\Delta k = sf(k) - \delta k$$

where

- Δk is a change in the capital per worker
- $sf(k)$ denotes investment per worker = **actual investment**
- δk denotes depreciation per worker = **break-even investment**

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Dynamics

- if $sf(k) = \delta k \implies \Delta k = 0$ **steady state**
- if $sf(k) > \delta k \implies \Delta k > 0$ capital accumulation
- if $sf(k) < \delta k \implies \Delta k < 0$ capital losses (e.g. war)

Solow model

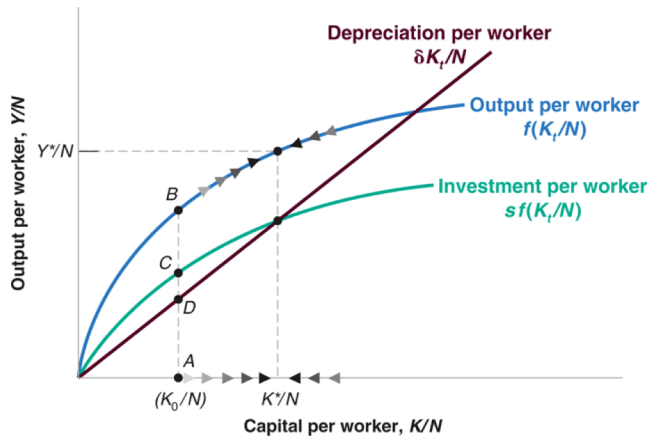


Figure: Caption

Solow model

In class, we will discuss how different variables affect the model dynamics. We will focus on:

- Changes in the capital per worker (k) $\implies$ movement along the curves
- Changes in the depreciation rate (δ) $\implies$ break-even investment line shifts
- Changes in the saving rate (s) $\implies$ real investment curve shifts

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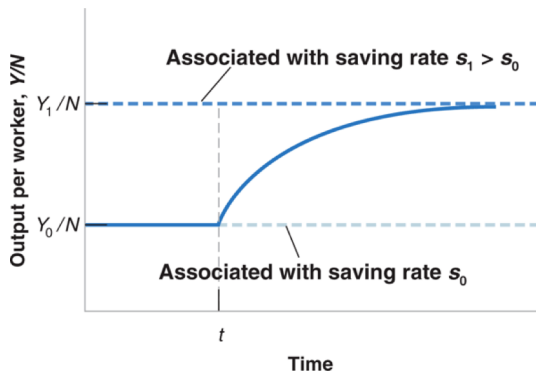
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- Saving rate **does NOT** affect **GROWTH RATE** of output per worker in the **long-run**. In the steady state, $\Delta k = 0$ and therefore also $\Delta y = 0$
- **BUT(!)** an increase in the saving rate will lead to **HIGHER** growth for some time **but not forever**

Convergence

An increase in the saving rate leads to a period of higher growth until output reaches its new higher steady-state level



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Golden rule level of capital:

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- **Graphically:** use the slope of the break-even investment line and find the point of tangency to the production function.
- **Mathematically:**

$$\begin{aligned}c &= f(k) - sf(k) \\ &= f(k) - \delta k\end{aligned}$$

$$\max_k c = \max_k (f(k) - \delta k)$$

$$\frac{dc}{dk} = 0$$

$$f'(k) - \delta = 0$$

Most economies: $s < s^{GOLD}$ Why?

- Trade-off: an increase in the saving rate leads to lower consumption for some time but higher consumption later.
- Recall: we are in the long-run: “later” can mean higher consumption in a couple of decades (next generations)